

STATIC GK — Science & Technology (Space)

Batch 13 of 14 · Standalone PYQ-book-only batch — per explicit scoping decision, this bucket has no independent raw-extraction reference-book backing (the raw extraction's own Computer chapter, used in Batch 12, is classic computer-fundamentals content and does not cover space technology) · Source: PYQ bucket "07 Science And Technology" (155 Qs, overwhelmingly ISRO/space-agency current affairs, 2023–2025) · ★ = tested in a real past SSC paper, cited directly from the PYQ book's own explanations (no book/PYQ cross-check possible here — see the HOOK box) · coverage note: this bucket contains dozens of near-duplicate questions re-testing the same handful of facts (e.g., Aditya-L1's L1 location, PSLV/GSLV basics) under different framings/Assertion-Reason wrappers — consolidated to one entry per distinct fact, same practice as Sports (Batch 5) · 4 questions in this bucket (UN Quantum Year 2025, Gemini AI, AI in healthcare) duplicate facts already cited in Batch 12 §6 and are not re-cited here

HOOK Unlike every other Static GK batch, this one carries no independent book source to fact-check against — the entire page is built from the PYQ book's own explanatory notes, which is why every ★ here cites the PYQ book itself rather than a Lucent/Parmar page. Treat every fact below as current-affairs-grade (accurate as printed in the PYQ book at compile time, subject to change as ISRO's programme progresses) rather than settled textbook fact.

1 Orbits & Satellite Mechanics

Item	Fact
Low Earth Orbit (LEO) speed ★	At 800 km altitude (Sun-Synchronous Orbit), a satellite travels at ≈ 7.5 km/s ★; speed decreases as altitude increases
Minimum satellite altitude ★	Satellites generally don't fly below 180 km ★ — denser atmosphere below causes drag/burn-up
Geostationary Orbit (GEO) ★	Altitude ≈35,786 km; orbital speed ≈ 3.07 km/s ★, matching Earth's rotation (24-hour period) — hence appears stationary over one location ★, used for continuous cloud/weather monitoring and as a "TV tower in the sky" at ≈ 36,000 km ★
Medium Earth Orbit (MEO) ★	Satellites in MEO do follow defined orbital paths (NOT arbitrary) ★; MEO (≈2,000–35,786 km) is used by navigation satellites — GPS, GLONASS, Galileo ★
International Space Station (ISS) ★	Completes ≈ 16 revolutions/day ★ (one orbit every ≈90 minutes, Low Earth Orbit)
Sun-Synchronous Orbit (SSO)	A satellite crosses the same local solar time on every pass — used for remote-sensing/Earth-observation satellites like NISAR
Artificial satellite examples ★	Sputnik-1 (world's first, USSR, 1957) and India's GSAT-1 are both correct examples of artificial satellites ★
Remote sensing satellites ★	Used for both drought assessment AND crop-yield estimation ★ (via rainfall/soil-moisture/NDVI data) — not one or the other
Launch vehicle physics ★	A satellite launch vehicle works on Newton's Third Law of Motion ★ (action-reaction; expelled gases push the rocket up)

2 ISRO Launch Vehicles

Vehicle	Fact
PSLV ★	= Polar Satellite Launch Vehicle ★; ISRO's most reliable "workhorse" — launched Chandrayaan-1 (2008), Mangalyaan (2013), Aditya-L1 (2023), 104 satellites in one mission (PSLV-C37, 15 Feb 2017 — a world record then) ★; versatile — places satellites in Polar/SSO, GTO, and LEO ★; no cryogenic stage (that's GSLV/LVM3) ★
GSLV ★	"G" = Geosynchronous ★; places satellites into geosynchronous orbit; GSLV-D1 (first developmental flight, Sriharikota, 18 April 2001) had three stages ★
GSLV Mk-III / LVM3 ★	Now officially called LVM3 ★ (Launch Vehicle Mark-3); India's heavy-lift vehicle; used for Chandrayaan-2, Chandrayaan-3, and the human-rated HLVM3 (Human-rated LVM3) for Gaganyaan ★

SSLV ★	Small Satellite Launch Vehicle — distinct because it enables on-demand, low-cost launches ★ , assembled in ≈72 hours vs. 60–90 days for PSLV/GSLV
RLV-TD / RLV-LEX ★	Reusable Launch Vehicle-Technology Demonstrator, ISRO's reusable-rocket testbed; "Pushpak" vehicle released from a Chinook helicopter and performed autonomous runway landing (RLV-LEX-02, cross-range corrections; RLV-LEX3, 23 June 2024, landed at >320 km/h) ★; RLV programme phases: HEX, LEX, REX, SPEX ★
Next Generation Launch Vehicle (NGLV) ★	Also called ' Soorya '/' Soham ' ★; approved by the Union Cabinet to support the Bharatiya Antariksh Station and a crewed Moon landing by 2040 ★; key feature = reusable booster with a semi-cryogenic (LOX-Methane) engine ★ — cleaner combustion, lower refurbishment cost than kerosene
Cryogenic engine — why hydrogen ★	Hydrogen has the highest energy density per kilogram ★ — more energy for less weight
Third launch pad, Sriharikota ★	Proposed to handle rising launch frequency for SSLV, Gaganyaan, and commercial payloads ★ (not a data-relay hub)

3 Moon Missions

Mission	Fact
Chandrayaan-1 ★	India's first lunar mission, launched 22 Oct 2008 (PSLV-C11, Sriharikota) ★; its M3 (Moon Mineralogy Mapper) payload (NASA) discovered water/hydroxyl on the lunar surface ★ — first confirmation of lunar hydration ★, overturning the "completely dry Moon" view
Chandrayaan-2 ★	Launched 2019 by GSLV Mk-III/LVM3 ★; orbiter + lander Vikram + rover Pragyan ★ (in that order); achieved detailed lunar South Pole mapping via the orbiter after the lander's hard landing
Chandrayaan-3 ★	Key objective = demonstrate safe soft landing and rover operations ★ on the Moon (3 goals: soft landing, rover ops, in-situ experiments); lander Vikram added a Lander Hazard Detection and Avoidance Camera ★ (LHDAC) vs. Chandrayaan-2; discovered a buried crater near the South Pole, ≈160 km diameter and ≈ 4.4 km deep ★; the RAMBHA-LP payload studies lunar plasma ★ (electron density/temperature); the Laser Retroreflector Array (LRA) onboard was supplied by NASA ★
LUPEX / Chandrayaan-4 ★	Lunar Polar Exploration Mission — joint ISRO-JAXA mission ★ to study water-ice quantity/quality/purity at the Moon's permanently-dark South Pole regions

4 Mars & Venus Missions

Mission	Fact
Mangalyaan / Mars Orbiter Mission (MOM) ★	India's first interplanetary mission ★ , launched 5 Nov 2013 via PSLV-C25 ★ from Satish Dhawan Space Centre; entered Martian orbit 24 Sept 2014 — India became the first country to reach Mars on its first attempt ★ (also the first mission in the world to do so on a first attempt)
Shukrayaan-1 (Venus Orbiter Mission) ★	Aims to map Venus's surface and study its atmosphere ★, particularly airglow and atmospheric dust ★; planned launch March 2028 ★

5 Solar Missions — Aditya-L1

Item	Fact
Aditya-L1 — basics ★	India's first solar mission ★, launched 2 Sept 2023 (PSLV-C57, Sriharikota) ★; placed in a halo orbit around the Sun-Earth Lagrange Point L1 ★ (≈1.5 million km from Earth, ≈1% of the Earth-Sun distance); took 127 days ★ to reach L1, entered halo orbit 6 Jan 2024; studies the Sun's photosphere, chromosphere, and corona ★ using visual/X-ray/UV instruments — NOT radar waves ★; does not orbit the Sun directly, and is not near the Moon ★

Why L1? ★	Lagrange points are where Sun-Earth gravity balances, letting a satellite stay stable with minimal fuel ★ — enables continuous, uninterrupted solar observation without eclipses
Aditya-L1 payloads (7 total) ★	VELC = corona imaging ★; SUIT = UV imaging of photosphere/chromosphere ★; SOLEXS = soft X-ray spectrometer (solar flares/corona heating) ★; ASPEX (incl. SWIS , Solar Wind Ion Spectrometer) = solar wind protons/alpha particles ★
Space-weather significance ★	Aditya-L1's corona/solar-wind data improves space-weather forecasting to protect satellites and power systems ★ from solar storms/CMEs
Other solar missions (non-ISRO) ★	Parker Solar Probe = NASA ★, first spacecraft into solar orbit; SOHO (Solar and Heliospheric Observatory) = joint ESA+NASA, Dec 1995, studies the Sun's interior/corona/solar wind from L1 ★

6 Human Spaceflight — Gaganyaan & Bharatiya Antariksh Station

Item	Fact
Gaganyaan ★	India's first human spaceflight mission; sends 3 astronauts ★ ("Vyomanauts"/"Gagan-yatris" — Prashant Nair, Ajit Krishnan, Angad Pratap, Shubhanshu Shukla named) into LEO at 400 km ★ altitude for 3 days, brought back safely; uses the human-rated HLVM3 ★; would make India the 4th country with human spaceflight capability after USA, Russia, China ★
Crew Escape System (CES) ★	Pulls the crew module away from the rocket to safety during a launch emergency ★; validated by the TV-D1 test flight (abort motor separated the crew module, parachute landing in the Bay of Bengal) ★
Bharatiya Antariksh Station (BAS) ★	India's first space station, in LEO ★, proposed altitude ≈400-450 km ★; first module (BAS-1) targeted for 2028 ★, fully operational by 2035 ★ (per ISRO's "Vision 2047"); significance = microgravity research, biotech experiments, long-duration spaceflight, Earth observation for disaster monitoring ★
Vision 2047 targets ★	BAS in orbit by 2035 ; first Indian on the Moon (indigenous technology) by 2040 ★
Space docking (SpaDeX) ★	India became the 4th country to demonstrate space docking (after USA, Russia, China) ★, via the SpaDeX mission, achieved 16 January 2025 ★; crucial for BAS construction and future lunar-sample-return missions ★

7 ISRO Satellites & Earth-Observation

Satellite	Fact
INSAT ★	Indian National Satellite System, est. 1983 ★; multipurpose — TV/radio broadcast, telecom, weather forecasting, disaster warning ★; largest domestic communication satellite system in the Asia-Pacific region; INSAT-3D/3DS = advanced meteorological satellites (cyclone/rainfall/climate monitoring) ★
INCOSPAR ★	"N" = National ★ (Indian National Committee for Space Research); constituted 1962 under Vikram Sarabhai; predecessor of ISRO
Cartosat-3 ★	High-resolution (25 cm) civilian remote-sensing satellite, launched 27 Nov 2019 via PSLV-C47 ★ — monitors agriculture/forestry/coastal resources ★
AstroSat ★	India's first dedicated multi-wavelength space observatory ★ (UV, optical, and X-ray simultaneously — unique among world satellites), launched Sept 2015 via PSLV-C30 ★; studies black holes, neutron stars, active galactic nuclei
NISAR ★	Joint NASA-ISRO Synthetic Aperture Radar mission ★; uses radar imaging ★ with dual-frequency L-band (NASA) and S-band (ISRO) ★; orbits in Sun-synchronous/polar orbit ★; revisits every 12 days ★ for land-deformation/earthquake monitoring; GS2 stage flagged off March 2025 ★
GSAT-9 (South Asia Satellite) ★	Gift to SAARC nations under PM Modi's "Neighbourhood First" policy ★, launched via GSLV-F09, 5 May 2017

GSAT-20 (GSAT-N2) ★	Delivers high-speed Ka-band broadband across India, esp. Andaman-Nicobar/Lakshadweep ★ (≈48 Gbps)
NVS-02 ★	Navigation satellite strengthening NavIC (NavIC = Navigation with Indian Constellation) ★ — precise Position/Velocity/Time services
DISHA ★	Twin-satellite mission studying Earth's upper atmosphere — Ionosphere and Thermosphere ★, and space weather
SCATSAT-1 ★	Scatterometer for ocean winds/weather/cyclone prediction ★, launched 26 Sept 2016 (PSLV-C35)
Trishna ★	Joint ISRO- France(CNES) ★ satellite for high-resolution thermal infrared imaging (Land Surface Temperature)
Megha-Tropiques ★	Joint India-France mission studying the water cycle/energy exchange in tropical regions ★ — helps predict monsoons
IDRSS ★	Indian Data Relay Satellite System — enables real-time communication with LEO satellites/astronauts ★
XPoSat ★	Studies black holes/neutron stars via X-ray polarimetry; POLIX measures X-ray polarization (8–30 keV) ★; XSPECT does X-ray spectroscopy/timing (not polarization) ★
DS-SAR (PSLV-C56, 30 July 2023) ★	SAR imaging satellite for Singapore's government/clients ★

8 ISRO — Institutions, Programme & Private Sector

Item	Fact
ISRO ★	Established 1969 ★, HQ Bengaluru ★
Satish Dhawan Space Centre (SDSC-SHAR) ★	ISRO's main spaceport, Sriharikota , Andhra Pradesh ★ — houses launch pads, Vehicle Assembly Building, Mission Control Centre; all major missions (Chandrayaan, Mangalyaan, Aditya-L1) launched from here
Vikram Sarabhai Space Centre (VSSC) ★	Kerala; designs and builds launch vehicles ★ (PSLV, GSLV, LVM3) — not a launch site
YUVIKA ★	ISRO's "Young Scientist Programme" ("Catch Them Young") ★ — introduces school students to space science
Space Situational Awareness (SSA) ★	Monitors space debris and protects Indian space assets ★, via radar/optical telescope network
Skyroot Aerospace ★	First private Indian company to launch a rocket into space ★ (18 Nov 2022, Sriharikota)
RHUMI-1 ★	India's first reusable hybrid rocket ★, by Tamil Nadu startups Space Zone India/Martin Group ★, launched 24 Aug 2024 from Mahabalipuram via a mobile launcher
QpiAI ★	Indian startup that built India's first full-stack 25-qubit quantum computer ★, under the National Quantum Mission
MATSYA 6000 ★	India's first manned scientific submersible, part of the Deep Ocean Mission ★ — carries 3 people to 6,000 m depth

9 Other Space Agencies & Missions

Mission/Agency	Fact
Europa Clipper ★	NASA mission to study Jupiter's moon Europa ★ — possible subsurface ocean
CryoSat ★	Operated by ESA ★ — studies polar ice thickness/climate change
James Webb Space Telescope (JWST) ★	Observes in near/mid- infrared ★ (statement 1 correct); located at Sun-Earth L2 ★ (NOT L1 — statement 2 in the tested set was actually false since it claimed L2 was correct but item framing tested distinguishing this); carries a coronagraph for exoplanet detection ★
Proba-3 (ESA) ★	Launched via India's PSLV-C59 , Oct 2024 ★ — tests precision "formation flying" between two satellites; a milestone in India's international space cooperation

Saturn's rings ★	Per NASA, disappearing via "Ring Rain" — estimated to vanish in ≈ 100 million years ★ (based on Cassini mission data)
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10 Miscellaneous Science & Technology

Item	Fact
Mission Mausam ★	Enhances weather-forecasting capability AND develops high-resolution monsoon-prediction models ★ (both objectives correct)
Bharat Forecast System ★	Adopted by IMD, 26 May 2025, developed by IITM Pune ★ (Indian Institute of Tropical Meteorology)
Beginning of India's space age ★	21 Nov 1963 ★ — first sounding rocket (Nike-Apache) launched from Thumba, near Thiruvananthapuram
Genome India Project ★	Maps genetic diversity of the Indian population ★ (Dept. of Biotechnology, Jan 2020, 10,000 genomes)
INSPIRE programme ★	Promotes science education ★ (DST, since 2008, ages 10–32); the INSPIRE-MANAK scheme takes student ideas via the INSPIRE Portal ★
GEMCOVAC-19 ★	India's first indigenous mRNA COVID-19 vaccine, by Gennova Biopharmaceuticals (Pune) with DBT/BIRAC support ★
PRITHVI Vigyan Mission ★	Unifies ocean/weather/geoscience observations AND supports tsunami/cyclone early-warning systems ★ — NOT launched by the Health Ministry (it's Earth Sciences) ★
Solar storms — effects ★	Can affect GPS signal accuracy, power transmission systems, submarine communication cables, AND pipeline infrastructure — all four ★
Prachand ★	India's first indigenous Light Combat Helicopter ★ , by HAL, for high-altitude combat zones

A.1 — PYQ citation index (representative)

Fact tested	Exam • Date • Shift
LEO satellite speed at 800 km ≈ 7.5 km/s	SSC CHSL TIER-I, 30/11/2025 (Shift-03)
Satellites don't fly below ≈ 180 km	SSC CPO TIER-I, 09/12/2025 (Shift-01)
Geostationary orbit speed ≈ 3.07 km/s, 36,000 km altitude	SSC CHSL TIER-I, 21/11/2025 (Shift-03)
ISS completes ≈ 16 revolutions/day	SSC CHSL TIER-I, 21/11/2025 (Shift-02)
PSLV = Polar Satellite Launch Vehicle, workhorse	SSC CGL TIER-I, 22/09/2025 (Shift-03)
GSLV "G" = Geosynchronous	SSC CHSL TIER-I, 27/11/2025 (Shift-02)
GSLV Mk-III = LVM3	SSC CHSL TIER-I, 26/11/2025 (Shift-02)
SSLV on-demand low-cost launches, 72-hour turnaround	SSC CGL TIER-I, 24/09/2025 (Shift-03)
RLV-TD/Pushpak autonomous landing tests	SSC PHASE XIII 29/07/2025 (Shift-03); SSC CPO TIER-I, 11/12/2025 (Shift-02)
NGLV = Soorya/Soham, LOX-Methane engine	SSC Stenographer 08/08/2025 (Shift-01); SSC CPO TIER-I, 10/12/2025 (Shift-01)
Chandrayaan-1 discovered lunar water/hydration	SSC CGL TIER-I, 19/09/2025 (Shift-02)
Chandrayaan-2 lander Vikram, rover Pragyan	SSC CHSL TIER-I, 17/11/2025 (Shift-03)
Chandrayaan-3 objective = safe landing/rover ops; LHDAC; buried crater 4.4 km deep; RAMBHA-LP=plasma; LRA by NASA	SSC Stenographer 06/08/2025 (Shift-02); SSC CGL TIER-I, 26/09/2025 (Shift-02); SSC CPO TIER-I, 09/12/2025 (Shift-01 & 12/12/2025 Shift-01); SSC CGL TIER-I, 15/09/2025 (Shift-02)
LUPEX = Chandrayaan-4, ISRO-JAXA	SSC Stenographer 06/08/2025 (Shift-01)
Mangalyaan first interplanetary mission, PSLV-C25, first-attempt Mars entry	SSC CHSL TIER-I, 25/11/2025 (Shift-02); SSC CGL TIER-I, 17/09/2025 (Shift-02)
Shukrayaan-1 studies Venus airglow/atmospheric dust, 2028 launch	SSC PHASE XIII 29/07/2025 (Shift-01); SSC PHASE XIII 25/07/2025 (Shift-02)

Aditya-L1 basics: L1, 127 days, halo orbit, corona study	SSC CHSL TIER-I, 17/11/2025 (Shift-03); SSC CGL TIER-I, 25/09/2025 (Shift-02); SSC PHASE XIII 01/08/2025 (Shift-02)
Aditya-L1 payloads: VELC/SUIT/SOLEXS/ASPEX	SSC PHASE XIII 24/07/2025 (Shift-03); SSC CGL TIER-I, 15/09/2025 (Shift-03)
Gaganyaan: 3 astronauts, 400 km, HLVM3	SSC CGL TIER-I, 14/10/2025 (Shift-02); SSC CGL TIER-I, 16/09/2025 (Shift-01)
Crew Escape System validated by TV-D1	SSC CPO TIER-I, 10/12/2025 (Shift-02)
BAS: LEO, 400-450 km, BAS-1 by 2028, full by 2035	SSC PHASE XIII 31/07/2025 (Shift-02); SSC CGL TIER-I, 12/09/2025 (Shift-03)
Vision 2047: BAS by 2035, Moon landing by 2040	SSC CGL TIER-I, 20/09/2025 (Shift-01)
SpaDeX docking, 4th country, 16 Jan 2025	SSC PHASE XIII 02/08/2025 (Shift-02); SSC CGL TIER-I, 13/09/2025 (Shift-03)
INSAT est. 1983, multipurpose	SSC CHSL TIER-I, 14/11/2025 (Shift-03)
INCOSPAR "N" = National	SSC CHSL TIER-I, 12/11/2025 (Shift-03)
Cartosat-3 25cm resolution	SSC PHASE XIII 01/08/2025 (Shift-01)
AstroSat first multi-wavelength observatory	SSC PHASE XIII 26/07/2025 (Shift-01); SSC Stenographer 08/08/2025 (Shift-03)
NISAR: L/S band, SSO/polar orbit, 12-day revisit	SSC CPO TIER-I, 10/12/2025 (Shift-01); SSC CGL TIER-I, 13/09/2025 (Shift-02 & 14/09/2025 Shift-03)
GSAT-9 gift to SAARC	SSC Stenographer 07/08/2025 (Shift-03)
ISRO est. 1969, HQ Bengaluru	SSC CGL TIER-I, 21/09/2025 (Shift-02)
Satish Dhawan Space Centre = Sriharikota	SSC CHSL TIER-I, 20/11/2025 (Shift-03)
VSSC designs/builds launch vehicles	SSC CGL TIER-I, 18/09/2025 (Shift-03)
Skyroot Aerospace first private Indian rocket launch	SSC PHASE XIII 31/07/2025 (Shift-03)
RHUMI-1 by Space Zone India	SSC CHSL TIER-I, 28/11/2025 (Shift-03); SSC Stenographer 06/08/2025 (Shift-02)
QpiAI 25-qubit quantum computer	SSC CHSL TIER-I, 13/11/2025 (Shift-03)
MATSYA 6000 = Deep Ocean Mission	SSC CHSL TIER-I, 24/11/2025 (Shift-03)
Europa Clipper studies Jupiter's moon Europa	SSC CHSL TIER-I, 29/11/2025 (Shift-03)
CryoSat operated by ESA	SSC CHSL TIER-I, 29/11/2025 (Shift-03)
JWST infrared, L2, coronagraph	SSC CGL TIER-I, 17/09/2025 (Shift-01)
Proba-3 launched via India's PSLV-C59	SSC CGL TIER-I, 14/09/2025 (Shift-02)
Saturn's rings disappearing (Ring Rain)	SSC CHSL TIER-I, 12/11/2025 (Shift-03)
Mission Mausam objectives	SSC CHSL TIER-I, 17/11/2025 (Shift-03)
Bharat Forecast System by IITM Pune	SSC CHSL TIER-I, 22/11/2025 (Shift-03)
India's space age began 1963, Thumba	SSC CHSL TIER-I, 20/11/2025 (Shift-03)
Genome India Project	SSC PHASE XIII 01/08/2025 (Shift-01)
GEMCOVAC-19 mRNA vaccine, Gennova	SSC PHASE XIII 28/07/2025 (Shift-01)
PRITHVI Vigyan Mission	SSC CGL TIER-I, 20/09/2025 (Shift-02)
Solar storm effects on GPS/power/cables/pipelines	SSC Stenographer 06/08/2025 (Shift-03)
Prachand = HAL Light Combat Helicopter	SSC CHSL TIER-I, 30/11/2025 (Shift-03)

Coverage: this index cites one representative exam/date/shift per distinct fact; the bucket's ~155 questions collapse to the ~90 distinct facts tabulated in sections 1–10 above (many questions re-test the same fact via Assertion-Reason, fill-in-the-blank, or "which of these is/isn't" framings). Duplicated with Batch 12 §6 and not re-cited here: UN's 2025 International Year of Quantum Science and Technology; AI improving healthcare via accurate diagnosis.

A.2 — PYQ-book-only facts

By construction, every fact in this batch is PYQ-book-only — there is no raw-extraction reference-book chapter on Indian space technology/ISRO in this Static GK compilation (Parmar has no Science & Technology chapter at all; Lucent's chapter 12 is purely about computers). All facts above are sourced to the PYQ book's own explanations and should be treated as current-affairs-grade rather than settled textbook fact, per the HOOK box.

A.3 — Doubts carried forward

None carried from a raw extraction (none exists for this topic). One internal inconsistency noted while compiling: the PYQ book's own JWST question (§9) frames "positioned at Sun-Earth Lagrange Point L2" as an incorrect statement in its stem numbering context, but its own explanation confirms JWST is indeed at L2 — the explanation is used here since it is the book's authoritative resolution of the question.